



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Fall 2024

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: 40

Duration: 2 Hours

Answer all questions. Figures are in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Read the following code and detect different types of errors with proper reasoning.

[8]

```
1. #include <studio.h>
2. #define PI=3.14159;
3. void calculateSquareAree(float side,);
4. floatt calculateRectangleArea(float length, float wide);
5. int main() {
6.     int choice;
7.     float side, length; width, base, leng; hight . n;
8.     char[n] radius;;

9.     print("Choose a shape to calculate its area:\n");
10.    printf('1. Square\n');
11.    printf("2. Rectangle/n");
12.    Printf("3. Triangle\n");

13.    scanf("%f", choice);
14.    switch (&choice) {
15.        case 1:{
16.            printf("Enter the side length of the square: ");
17.            scanf("%2f", &side);
18.            printf("Area of the square: %.2f\n",    calculateSquareArea(side));
19.            return 0;
20.        }
21.        case 2:{
22.            printf("Enter the length and width of the rectangle: ");
23.            scanf("%f %ld", &length, &width);
24.            printf("Area of the rectangle: %.2d\n", radius =    calculateRectangle(length, width));
25.            break(D;
26.
27.        case 'c':
28.            printf("Enter the base and height of the triangle: ");
29.            scanf("%f %f", &base; height);
30.            printf("Area of the triangle: %.2f\n",    calculateTriangleArea(base, height,n));
31.        }

32. void calculateTriangleArea(float base, float hieght){
33.     return 0.5 * base * height;
34. }
```

```
35.float calculateRectangleArea(float length, int w){
36.    return calculateRectangleArea(length--, w++);
37. }

38. int calculateSquareAree(float side) {
39.     int k = side * sides;
40. }
```

2. For the following grammars, write down the **FIRST** and **FOLLOW** sets for each nonterminal. Consider [4*3]
all the terminals are of single character/number/symbol.

$$\begin{aligned} \text{a) } A &\rightarrow aBi \mid bBCDiu \mid qBrC \mid DBCTu \mid \epsilon \\ B &\rightarrow 1BAi \mid 2CDu \mid 3ApBg \mid CD2 \mid \epsilon \\ C &\rightarrow CrDt \mid zDBl \mid BBCy2 \mid iABcDkB \mid \epsilon \\ D &\rightarrow 0D1B \mid 3x2 \mid DD2Ai \mid 3B2ACi \mid \epsilon \end{aligned} \quad \begin{aligned} \text{b) } S &\rightarrow BbNGla \mid NxGyl \mid 23NgS \mid \epsilon \\ B &\rightarrow 3GGS \mid BiBSr \mid rSBt \mid GGkNN \mid \epsilon \\ N &\rightarrow iBgNS \mid rBgB \mid olBGB \mid tGn \mid \epsilon \\ G &\rightarrow 3GS3 \mid GkBkl \mid 2BSi \mid 4BiB \mid \epsilon \end{aligned}$$

$$\begin{aligned} \text{c) } P &\rightarrow bPrQ \mid bRRt \mid RrQ \mid ab \\ Q &\rightarrow 3QRi \mid PQ25 \mid xRgPl \mid \epsilon \\ R &\rightarrow 1RQk \mid 3tQPo \mid RRYi \mid \epsilon \end{aligned}$$

3. **Left Factor** the following.

[5*2]

$$\text{(a) } P \rightarrow [2 + 3]i \mid [2234]y \mid 345 \mid 33iop \mid 33kip \mid 33iol \mid 33jle \mid 32ir \mid 322pl \mid 324 \mid 322yu \mid 345uit \mid 345uk \mid 34qio \mid 33ilp \mid 33ikol \mid 33kkpo \mid 33ki \mid 33k \mid [1yp] \mid [2yui] \mid 36hy \mid 4$$

$$\text{(b) } R \rightarrow \text{Artic} \mid \text{Arcani} \mid \text{Arcame} \mid \text{Arcanio} \mid \text{Arc Love} \mid \text{Arcaml} \mid \text{Artnet} \mid \text{Arcock} \mid \text{Arces} \mid \text{Arc is curve} \mid \text{Arcocips} \mid \text{Arcockeak} \mid \text{Burnish} \mid \text{Butter} \mid \text{Bur} \mid \text{Buter} \mid \text{Buteeo} \mid \text{Burri} \mid \text{Burnit} \mid \text{Burnnop} \mid \text{But} \mid \text{Burrk} \mid \text{Buteos} \mid \text{bute} \mid \text{butto} \mid \text{buttop} \mid \text{buttkr}$$

4. Eliminate immediate **left recursion** from the following grammar.

[5*2]

$$\begin{aligned} \text{(a) } S &\rightarrow Sa \mid aA \mid aB \mid SS \\ A &\rightarrow Ab \mid xBy \mid AAC \mid \epsilon \\ B &\rightarrow iAc \mid BA \mid cC \mid \epsilon \\ C &\rightarrow Cd \mid dA \mid CCaC \mid f \end{aligned} \quad \begin{aligned} \text{(b) } X &\rightarrow XZ \mid XXZN \mid MX/YX \mid /ZY \mid XNMc \\ Y &\rightarrow Y*bb \mid YX+Mc \mid Y/Md \mid Y-Ze \mid Y*Y*M \mid Y \mid YZYM \\ Z &\rightarrow ZZ+Y \mid Y+Zb \mid Z*X/M \mid (Z) \mid /ZM \mid Z+Mc \mid \epsilon \\ M &\rightarrow (M) \mid M+(X*Y) \mid M+b \mid N*Y+Z \mid M+ZgZ \mid M*bZ \mid Z+Mc \\ N &\rightarrow NN-Ze \mid Y*Y*N \mid Y \mid XY \mid NZYM \mid \epsilon \end{aligned}$$